New York University
Computer Science Department
Introduction to Software Development
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Homework 5

Deadline: *See NYUclasses for the deadline, 5% off per day after the deadline (3 days maximum).*

Please read the guidelines carefully to avoid receiving a zero on the HW:

- Please use the following naming convention "ExerciseX.java", where X is the number of the exercise.
- Apply this convention to all future homework assignments during the semester.
- Compile and run the program for each exercises.
- Make an archive (zip file or compressed file) with all the java files (the .java files NOT the .class files) and post it on NYU Classes (StudentName_Homework_X.zip)
- It is your responsibility to make sure that the Zip files have your actual files. You may send the file to yourself by email to double check before you upload on NYU classes.
- If the graders cannot open the file, you will receive a grade of zero.
- If you send the .class files instead of the .java files (source files) you will
- receive a zero.
- Cheating will be severely addressed with an immediate zero on the homework and a report to the academic advisor and the administration.
- You will automatically lose 50% of the points for an exercise if the program does not compile and run correctly
- Do not use eclipse or any IDE, only text editors and command line

Exercise#1 (10pts): (Reverse the numbers entered) Write a computer program that reads ten integers and display them in the reverse of the order in which they read.

Exercise#2 (20pts): Write a program that reads integers between 1 and 100 and counts the occurrences of each. Assume the input ends with 0.

Here is a sample run of the program:

Enter the integers between 1 and 100: 2 5 6 5 4 3 23 43 2 0

Output:

2 occurs 2 times, 3 occurs 1 time, 4 occurs 1 time,

5 occurs 2 times, 6 occurs 1 time, 23 occurs 1 time, 43 occurs 1 time

Exercise#3 (20pts): Go to https://cims.nyu.edu/~kapp/cs101/arrayders/

There are four challenges, 5 points each.

Create three java classes name each one as challenge1.java, challenge2.java, challenge3.java for the first three challenges. For the final challenge, include a screenshot after you successfully enter the vault by providing the passwords you collected from each of the three challenge.

Exercise#4 (20pts): Implement the following method that returns true if the array is already sorted in increasing order.

public static boolean **isSorted**(int[] list)

Write a test program that prompts the user to enter a list of int numbers and displays whether the list is sorted or not.

Here is a sample run. Note that the first number in the input indicates the number of elements in the list. This number is NOT part of the list.

e.g. Input: Enter list: 8 10 15 16 6 1 9 11 1

Output: The list is not sorted

e.g. Input: Enter list: 10 1 1 3 4 4 5 7 9 11 21

Output :The list is already sorted

Exercise#5 (30pts): Create a program that generate a random 7x7 array (1-100) and has two methods

a. **void sortArrayRows(int[][] arr)** method which sorts each row of the array individually, then print a well formed array

example on a 3x3 2D array:

```
input:  17  7  27      printed output:  7  17  27
        12  2  1                        1   2  12
        8   70 45                       8  45  70
```

b. **void sortArray(int[][] arr)** method which sorts the entire array ascending (from smallest to largest) then print a well formed array

example on a 3x3 2D array:

```
input:  17  7  27      printed output:   1  2   7
        12  2   1                        8  12 17
         8 70  45                       27  45 70
```